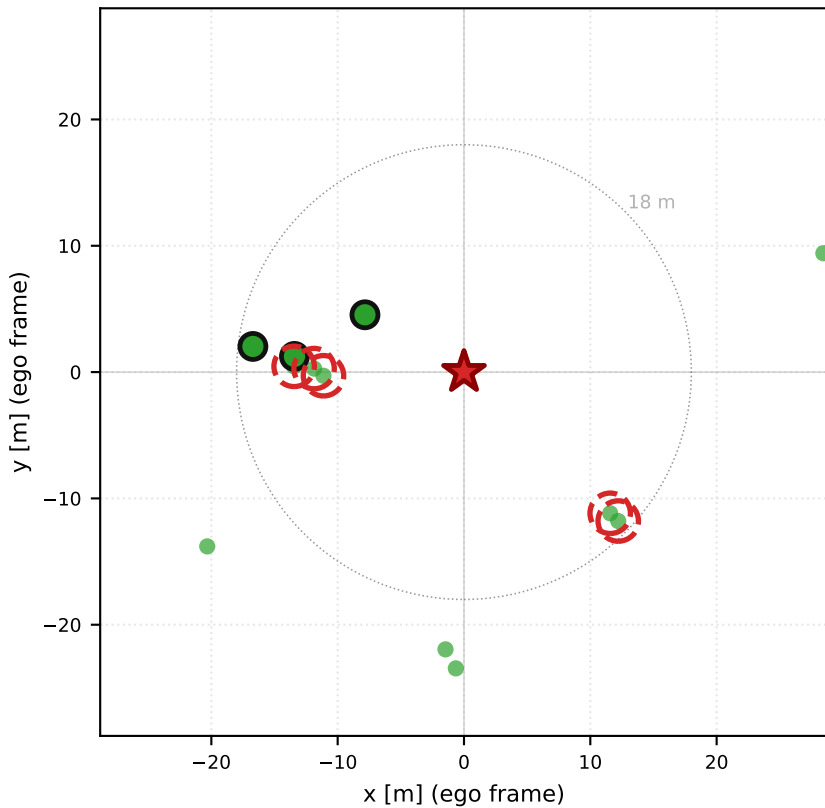
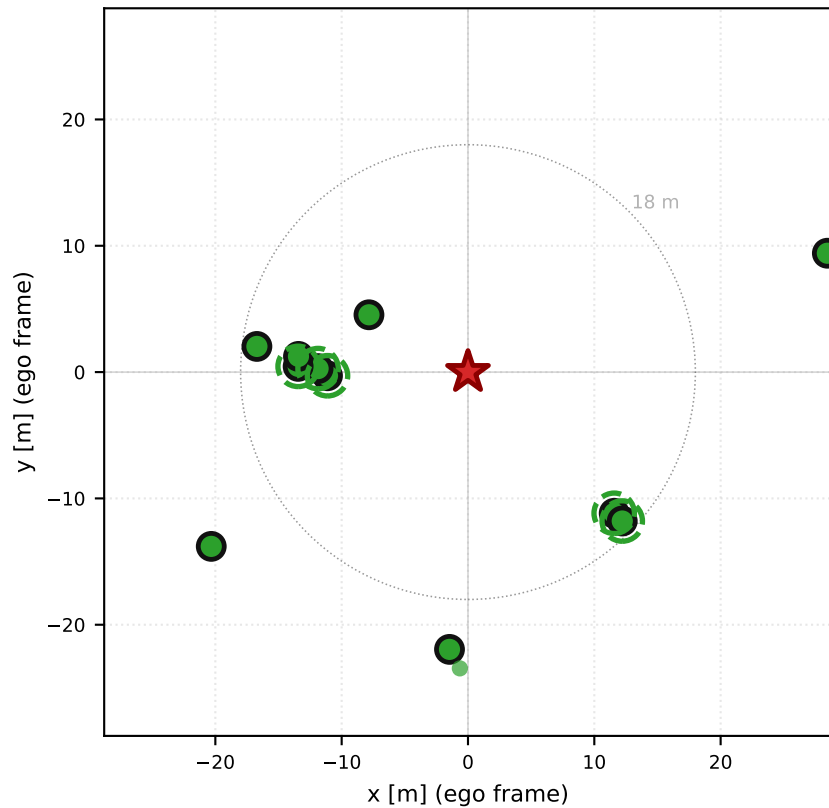


Naive shared top-k drops dynamic agents that typed-budget keeps (prediction-err ratio 12×)

Object+Relation-16 (naive, shared top-k)
3 / 8 near agents selected · pred err = 2.70 m



Decoupled typed-budget (Ours, 12 dyn + 4 rel)
8 / 8 near agents selected · pred err = 0.22 m



★ EGO ● VEH ● PED ● CYC ● selected by abstraction ● dropped by naive (kept by decoupled)